MIL-C-51404A
11 July 1978

SUPERSEDING
MIL-C-51404(EA)
14 May 1974

MILITARY SPECIFICATION

CRIMPING OUTFIT, HYDRAULIC, VOICEMITTER-OUTLET VALVE ASSEMBLY, M1

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers a voicemitter-outlet valve crimping device.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Steel, and Seals.
MMM-A-1617	-	Adhesive, Rubber Base, General Purpose.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation Packaging, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.

FSC 5180

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-B-131 - Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-G-13210 - Gaskets, Rubber.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

EDGEWOOD ARSENAL

E5-1-553 - Facepiece, Field Chemical-Biological Mask, M17A1.
E5-2-948 - Facepiece, Field Protective Mask, ABC-M17.
D77-1-30 - Crimping Outfit, Hydraulic, Voicemitter-Outlet Valve Assembly, M1.
E136-13-770 - Tester, Pull, Nosecup-Mask Assembly, Q162.
E136-42-750-2D - Tester, Leakage, Protective Mask, DOP, M14.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM3-5180-210-15 - Crimping Outfit, Hydraulic, Voicemitter-Outlet Valve Assembly, M1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components. All components of the crimping outfit shall conform to the specifications and drawings listed on Drawing D77-1-30.

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3.2 Manufacture and assembly. The crimping outfit shall be manufactured and assembled in accordance with Drawing D77-1-30.

3.3 Operation. The crimping outfit shall function properly and shall effectively crimp the voicemitter-outlet valve to faceblanks and nosecups when tested as specified in 4.2.4.

3.4 Leakage. Leakage of the crimped joint shall not exceed 0.003 percent dioctylphthalate (DOP) smoke penetration, before and after the pull test, when tested as specified in 4.2.4.2.

3.5 Pull test. There shall be no creeping or separation of the nosecup from the voicemitter-outlet valve crimped joint when tested as specified in 4.2.4.3.

3.6 Workmanship. The crimping outfit shall be properly assembled and free from rust, foreign matter, and damage such as cracks, dents, or other damage which could affect its intended use. This will include proper application of, or subsequent damage to, protective coatings such as paint or plating.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2 and section 5 for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the crimping outfits produced by one manufacturer, at one plant, using the same materials, components, and essentially the same manufacturing conditions.

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4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination. The crimping outfits shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For test. The crimping outfits shall be tested in accordance with 4.2.4. The acceptance number is zero.

4.2.3.3 Classification of defects.

(a) Crimping outfit, hydraulic, voicemitter-outlet valve assembly, M1 (Dwg D77-1-30).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Component missing or incorrect
102		Moving parts not operating properly
103		Hydraulic fluid leaking
104		Workmanship (3.6)

4.2.4 Tests.

4.2.4.1 Operation. Each crimping outfit shall be put into operation and ten new C29R1 voicemitter-outlet valves or C29R5 voicemitter-outlet valves shall be crimped into M17 or M17A1 mask facepieces in accordance with either Drawing E5-2-948 or E5-1-553, whichever is applicable.

4.2.4.2 Leakage. Each of the ten crimped joints (4.2.4.1) shall be tested for leakage with the M14 DOP Protective Mask Leakage Tester (E136-42-750-2D).

4.2.4.3 Pull test. Each of the ten crimped joints (4.2.4.1) shall be tested by applying a 1/4-inch, 10-second pull with the Q162 Nosecup-Mask Assembly Pull Tester (E136-13-770).

5. PACKAGING

5.1 Packaging, level A. All packages shall be of the minimum cube required to contain the items.

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5.1.1 Cleaning and drying. The crimping outfit, components, and repair parts thereto shall be cleaned and dried in accordance with the applicable requirements of MIL-P-116, using cleaning method C-1 and any drying method listed therein.

5.1.1.1 Preservation and packaging methods. Unless otherwise specified, preservatives including the application thereof and packaging methods cited herein shall be in conformance with the requirements of MIL-P-116.

5.1.1.2 Unit preservation and packaging.

5.1.1.3 Pressure gage assembly. The pressure gage and hardware shall be packaged in accordance with method IA-15. The gage shall be wrapped with cellulosic cushioning material conforming to type I, class B of PPP-C-843 and secured with pressure sensitive tape conforming to type IV, class 1 of PPP-T-60. The hardware shall be put in the same box with the pressure gage.

5.1.1.4 Pliers. Each pliers shall be coated with P-1 preservative.

5.1.1.5 Oil, hydraulic. The oil shall be packaged as specified in MIL-H-5606.

5.1.1.6 Gasket compound. The gasket forming compound shall be packaged in a tube (commercial type, 11 oz).

5.1.1.7 Pump handle. The pump handle, after being disconnected from the unit, shall be cleaned and coated with a type P-9 preservative compound and wrapped with greaseproof barrier material conforming to grade C of MIL-B-121, securing the wrap in place with pressure-sensitive tape conforming to type IV, class 1 of PPP-T-60.

5.1.1.8 Handle assembly. The handle assembly shall be coated with type P-9 preservative compound and wrapped with greaseproof barrier material conforming to grade C of MIL-B-121, securing the wrap in place with pressure-sensitive tape conforming to type IV, class 1 of PPP-T-60.

5.1.1.9 Technical publications. Technical publications TM3-5180-210-15 and the hydraulic press operator's and maintenance manual shall be packaged in accordance with method IC-3.

5.1.2 Intermediate packaging. Items packaged as specified in 5.1.1.3 through 5.1.1.6, shall be packed in a style RSC fiberboard box conforming to W6c of PPP-B-636, having an approximate inside dimensions of 11-1/8 by 5-1/8 by 12 inches. Cellulosic cushioning material conforming to type

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I, class B of PPP-C-843 shall be used to make a tight pack. The box shall be closed and sealed, including the manufacturer's joint, in accordance with the appendix to the box specification. Items packaged as specified in 5.1.1.7 and 5.1.1.8 shall be packed in a style RSC fiberboard box conforming to W6c, of PPP-B-636, having an approximate inside dimension of 5-1/4 by 4-1/4 by 30-1/2 inches. Cellulosic cushioning material conforming to type I, class B of PPP-C-843 shall be used to make a tight pack. The box shall be closed and sealed, including the manufacturer's joint, in accordance with the appendix to the box specification.

5.2 Packing.

5.2.1 Level A. The crimping outfit, together with the intermediate packages and technical publications, packaged as specified in 5.1, shall be packed in an overseas type, style A or B cleated plywood box conforming to PPP-B-601. The crimping outfit shall be anchored to the base of the box with four 1/4 inch diameter carriage bolts. The bottom of the box shall be fitted with nominal 2 by 4 inch wood skids positioned flatwise, parallel to the length of the box, and located in such a manner as to permit all bolts used in anchoring of the unit to the base of the box to pass through the center of the wood skids. The skids shall be notched on that surface that comes in contact with the box bottom, so as to permit the steel strapping to pass between the skids and the box bottom. Wood filler strips shall be used where required so as to take up the void between the skid and the bottom of the box. The intermediate package containing the pump handle, handle assembly, etc., shall be put in the vacant space provided in the chassis, around the area of the pump handle of the crimping unit and secured with pressure-sensitive tape conforming to Type IV, Class 1 of PPP-T-60. The remaining intermediate package shall be placed in the space provided underneath the lower die assembly. Fiberboard pads shall be used to fill the remaining area around the fiberboard box to make a tight pack. A minimum of one inch clearance shall be maintained between the inside surface of the top, sides, and ends of the box. Prior to packing the crimping outfit within the box, a flexible water vaporproof barrier material conforming to class 1 of MIL-B-131 and containing holes to permit the installation of the barrier material over the carriage bolts shall be placed over the crimping unit. The holes in the barrier material shall be gasketed on both sides of the material. The gaskets shall be 1/8 inch thick and conform to MIL-G-13210, and shall be attached to the barrier material with adhesive conforming to MMM-A-1617. The upper and lower die assembly shall be cleaned and coated with a P-9 preservative of MIL-P-116 and covered with greaseproof barrier material conforming to grade C of MIL-B-121 and secured with pressure-sensitive tape conforming to Type IV, class 1 of PPP-T-60. Two 1/2 inch diameter holes shall be bored in a block of wood to fit between the upper and lower

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die assembly. The block of wood and the guide mounted on the frame of the crimping outfit shall be wrapped with cellulosic cushioning material conforming to type I, class B of PPP-C-843. The pressure shall be adjusted to eliminate any movement of the die assembly while in storage or shipment. The release lever shall be rotated 180 degrees. A block of wood shall be placed between the release lever and the arm of the pump assembly and secured with pressure-sensitive tape conforming to type IV, class 1 of PPP-T-60. The packaged technical publications shall be secured to the frame of the crimping outfit and secured with pressure-sensitive tape conforming to type IV, class 1 of PPP-T-60. The box shall be closed and strapped in conformance with the applicable container specification and the appendix thereto, using galvanized strapping conforming to QQ-S-781.

5.2.2 Packing, commercial. The crimping outfit assembly and components, packaged as specified in 5.1, shall be commercially packed in accordance with MIL-STD-1188.

5.3 Marking, level A and commercial. In addition to any special marking specified in the contract or purchase order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188 as applicable.

6. NOTES

6.1 Intended use. The M1 crimping outfit is intended to be used for crimping both the C29R1 voicemitter-outlet valve assembly into M17 masks and the C29R5 voicemitter-outlet valve assembly into M17A1 masks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

6.3 Masks and voicemitters. Either the M17 masks and C29R1 voicemitter-outlet valve assemblies or M17A1 masks and C29R5 voicemitter-outlet valve assemblies required for testing will be furnished by the Government and should be obtained as directed by the contracting officer.

Custodian:

Army - EA
Air Force - 99

Preparing activity:

Army - EA

Project No. 5180-0119

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-51404A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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